Agenda

- Sampling
- Statistic
- Plots

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Example 1 Quality Control

- 1000 rods are produced to meet the specification 0.45 ± 0.02 cm
- No time and no resource to check all 1000 rods.
- What to do?

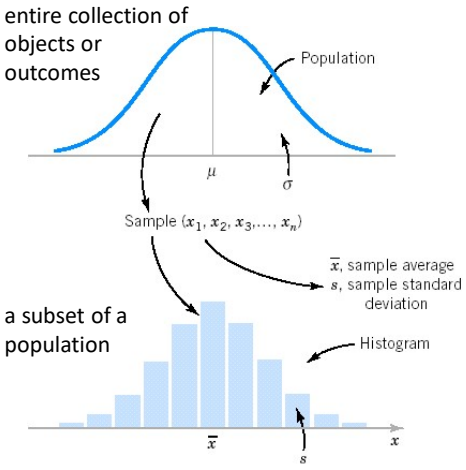
Here are some relevant questions:

1. How large is a typical difference between the sample proportion and the population proportion for this kind of sample?
2. What interval gives a good estimate of the percentage of acceptable rods in the population with reasonable certainty?
3. How certain can the engineer be that at least 90% of the rods are good?

Statistics can help us to address questions like these.

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How to link the two worlds? Population and Sample



- A **simple random sample (S RS)** of size *n* : a sample chosen by a method in which each collection of *n* population items is equally likely to comprise the sample, just as in the lottery.

- The random variables $X_1,\ldots,X_n$ are said to **simple random sample** of size *n* if:
  (1) X_i's are independent
  (2) X_i's are identical

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Independence

- The items in a sample are said to be **independent** if knowing the values of some of the items does not help to predict the values of the others.
  → choosing one item for the sample has no effect on the choice of other items
- Items in a simple random sample may be treated as independent in most cases encountered in practice, unless the population is finite, and the sample comprises a substantial fraction (more than 5%) of the population.

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How to draw a random sample?

- A professor's question: average height of students at a university?
- Population?
- How to choose a sample of size of 100?

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No: 178



What can be wrong?



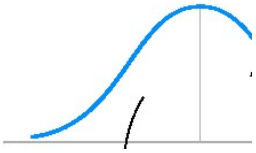
15

A Random Sample

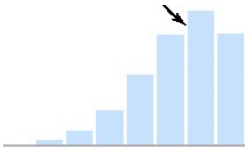
Definition: The random variables $X_1,\ldots,X_n$ are said to form a (simple) random sample of size n if

(1) X_i's are independent r.v.s,

(2) X_i's have the same probability distribution



An additional issue is on whether a random sample is representative of the underlying population or not.

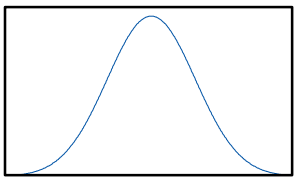


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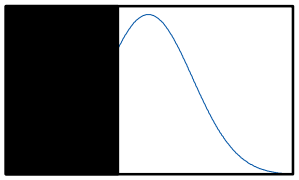
A Random Sample

- Should be representative of the whole population.
- How reliable is the data set? Does it represent the whole population?

Population



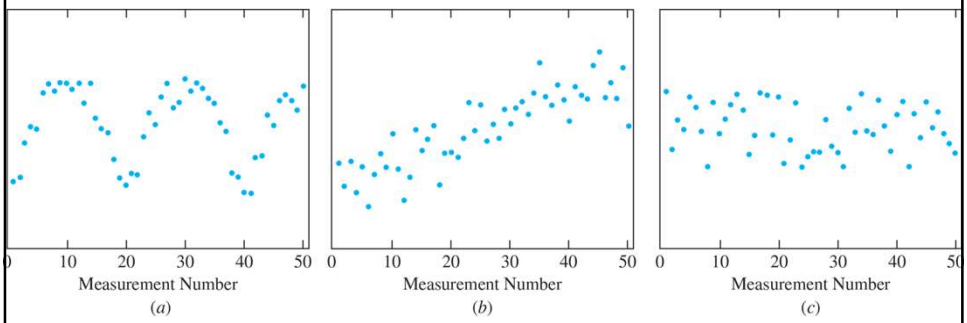
Population of people who are poorer.



Violates in independence assumption.

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Which One(s) Random Sample?



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Simple Random Sampling

A SRS is not guaranteed to reflect the population perfectly.

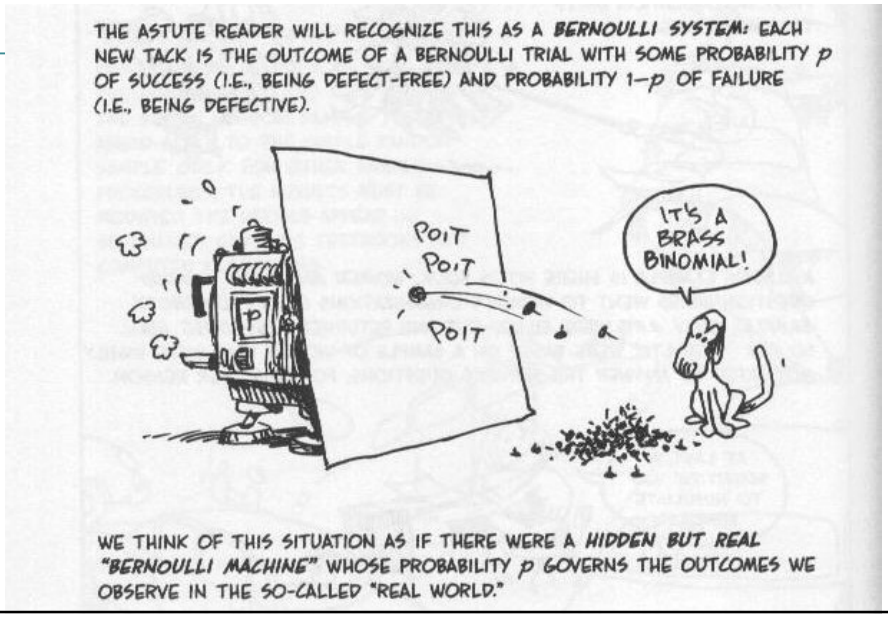
SRS's always differ in some ways from each other; occasionally a sample is substantially different from the population.

Two different samples from the same population will vary from each other as well.

- This phenomenon is known as **sampling variation**.

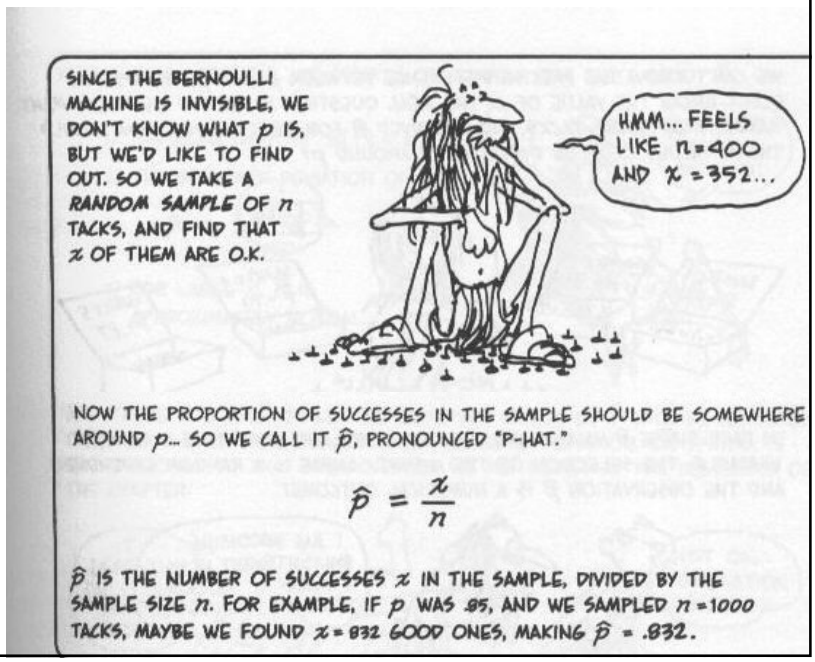
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Tack Machine



20

Estimate probability of non-defective?



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If we took many samples of 1000 tacks:



This phenomenon is known as sampling variation.

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Example 1 Quality Control

- 1000 rods are produced to meet the specification 0.45 ± 0.02 cm
- No time and resource to check all 1000 rods.
- 50 rods are checked → 46 of them met specification → 92%
- What can we say about the population of 1000 rods?

Sampling variation → the true proportion of rods that meet specification will be close to the sample proportion, or 92%, but not exactly equal.

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A conceptual population

All of the values that might possibly have been observed from a population.

- For example, a geologist weighs a rock several times on a sensitive scale. Each time, the scale gives a slightly different reading.
- Here the population is conceptual. It consists of all the readings that the scale could in principle produce.

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A sample of convenience

A sample that is obtained in some convenient way and is not drawn by a well-defined random method.

Things to consider:

- Convenience samples may differ **systematically** in some way from the population.
- Thus, they should only be used when it is not feasible to draw a random sample.
- If you do have a convenience sample, think carefully about the ways in which the sample might differ systematically from the population.

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How to draw a sample of convenience?

- A professor's question: average height of students at a university?
- Population?
- How to choose a sample of size of 100?

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Other Sampling Methods

Samples other than simple random samples can be useful in various situations.

These include (but are not limited to):

- Weighted sampling.
- Stratified random sampling.
- Cluster sampling.

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Types of Experiments

Many types of experiments can be used to generate data.

One-sample experiments involve only one population of interest with a single sample drawn from it.

Multi-sample experiments involve two or more populations of interest with a single sample drawn from each population.

- In many multi-sample experiments, the populations can be distinguished from one another by varying one or more **factors** that may affect the outcome of interest.

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Types of Data

When a numerical quantity designating how much or how many is assigned to each item in the sample, the resulting set of values is **numerical or quantitative**.

- Height (in centimeters).
- Weight (in kilograms).
- Age (in years).

When sample items are placed into categories and category names are assigned to the sample items, the data are **categorical or qualitative**.

- Hair color.
- Country of origin.
- Zip code.

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Controlled Experiments and Observational Studies

A **controlled experiment** is one in which the experimenter controls the values of the factors.

- When designed and conducted properly, controlled experiments can produce reliable information about cause-and-effect relationships between factors and response.

An **observational study** is one where the experimenter simply observes the factors as they are, without having any control over them.

- Observational studies are not nearly as good as controlled experiments for obtaining reliable conclusions regarding cause and effect.

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Statistic

- Let $X_1, \ldots, X_n$ be a (simple) random sample.
- A statistic is any function of the observations in a random sample.
- A statistic is a random variable, so it has its own probability distribution (generally called sampling distribution).

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Examples of statistics

Let $X_1, \ldots, X_n$ be a random sample.

- Sample mean
- Sample standard deviation
- Minimum value of the data
- Maximum value of the data
- Mode
- Median
- Quartiles
- …

How about their distributions?

32

Measures of central tendency

- Do we mean by “average”?

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A shoe manufacturing business

- 3 partners: CEO, production manager, sales manager
- 5 engineers
- 40 workers
- Average salary in this business?

CEO	40000
PM	30000
SM	30000
E1	10000
E2	10000
E3	10000
E4	10000
E5	10000

W1	3000
W2	3000
W3	3000
W4	3000
W5	3000
W6	3000
W7	3000
W8	3000
W9	2500
W10	2500

W11	2500
W12	2500
W13	2500
W14	2500
W15	2500
W16	2500
W17	2500
W18	2500
W19	2500
W20	2500

W21	2000
W22	2000
W23	2000
W24	2000
W25	2000
W26	2000
W27	2000
W28	2000
W29	2000
W30	2000

W31	2000
W32	2000
W33	2000
W34	2000
W35	2000
W36	2000
W37	2000
W38	2000
W39	2000
W40	2000

34

Definitions: Measures of central tendencies

- Let $X_1, \ldots, X_n$ be a random sample.
- Sample mean is the arithmetic average of all observations.

$$\bar{x} = \frac{x_1 + x_2 + \cdots + x_n}{n} = \frac{\sum_{i=1}^{n} x_i}{n}$$

- Median divides the data into two equal parts, so that the number of observations below and above the median are equal.
- Mode is the value that is observed most frequently.

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A shoe manufacturing business

- 3 partners: CEO, production manager, sales manager
- 5 engineers
- 40 workers
- Average salary in this business?



Mean	5083
Median	2500
Mode	2000

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What kind of average?

- It makes a difference!

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A shoe manufacturing business

- 3 partners: CEO, production manager, sales manager
- 5 engineers
- 40 workers
- Average salary in this business?

What if 40000 is changed to 20000? What are the new values of mean, median and mode?



Mean	5083
Median	2500
Mode	2000

Mean	4667
Median	2500
Mode	2000

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Changes in these measures

- Mean is very sensitive to very high and very low values.
- Median is much more stable.

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Sample mean

Eight prototype units for engine connectors are produced. Their pull-off forces are measured. If the pull-off force is too low, the connector may fail when it is installed in an engine.

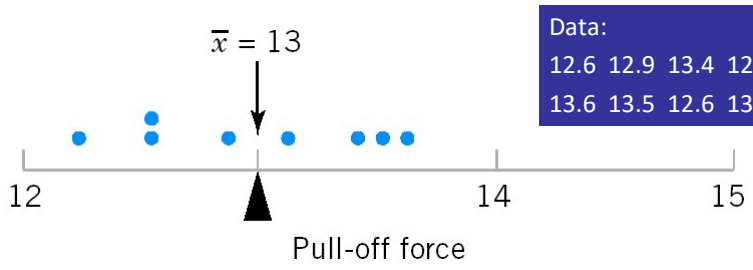
Data:
12.6 12.9 13.4 12.3
13.6 13.5 12.6 13.1

$$\bar{x} = \frac{x_1 + x_2 + \cdots + x_n}{n} = \frac{\sum_{i=1}^{8} x_i}{8} = \frac{12.6 + 12.9 + \cdots + 13.1}{8}$$

$$= \frac{104}{8} = 13.0 \text{ pounds}$$

40

Figure



Data:
12.6 12.9 13.4 12.3
13.6 13.5 12.6 13.1

The sample mean as a balance point for a system of weights.

41

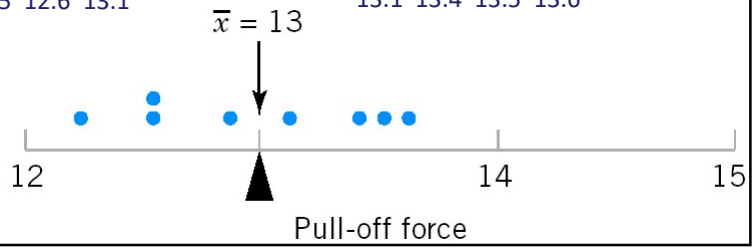
Definitions:

Measures of central tendencies

- Mode is the value that is observed most frequently.
- Median divides the data into two equal parts.

Data:
12.6 12.9 13.4 12.3
13.6 13.5 12.6 13.1

Sorted data:
12.3 12.6 12.6 12.9
13.1 13.4 13.5 13.6



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Finding the Sample Median

Order the *n* data points from smallest to largest. Then

- If *n* is odd, the sample median is the number in position $\frac{n+1}{2}$.
- If *n* is even, the sample median is the average of the numbers in positions $\frac{n}{2}$ and $\frac{n}{2}+1$.

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What kind of average?

- It makes a difference in some cases, but not for all!
- When the underlying distribution is symmetric, we expect that all three measures are close to each other.
- When the underlying distribution is ***skewed***, we expect them to be very different.

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Fallacy of Averages

- Did you hear about the statistician who had his head in an oven and his feet in a bucket of ice?
- When asked how he felt:

  He replied, ***"On the average I feel just fine."***

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What else will you look at in the data set?

Variability?

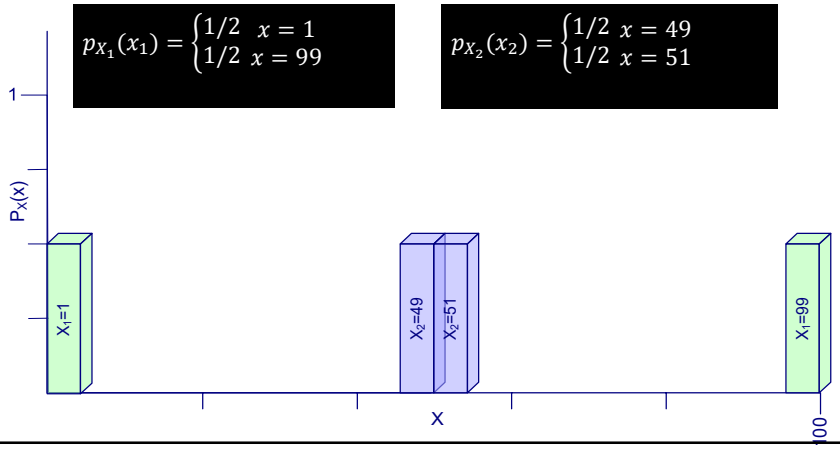
46

What is variability?

- Compare the two random variables X_1 and X_2:

$$p_{X_1}(x_1) = \begin{cases} 1/2 & x = 1 \\ 1/2 & x = 99 \end{cases}$$

$$p_{X_2}(x_2) = \begin{cases} 1/2 & x = 49 \\ 1/2 & x = 51 \end{cases}$$



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Population variance

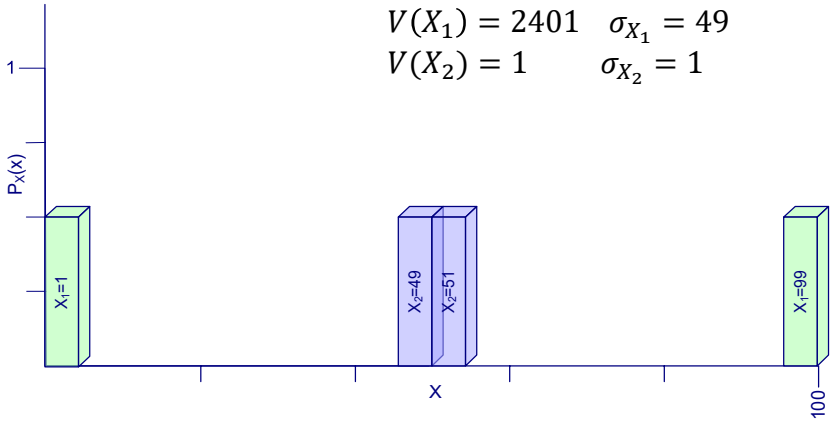
$$V(X) = E((X-\mu)^2) = \begin{cases} \sum_{\forall x} (x-\mu)^2 \, p(x) & X: discrete \\ \int_x (x-\mu)^2 f(x) dx & X: continuous \end{cases}$$

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What does variance measure?

- Variance measures how spread the random variables are.

$$V(X_1) = 2401 \quad \sigma_{X_1} = 49$$
$$V(X_2) = 1 \quad \sigma_{X_2} = 1$$



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Sample Variance and Standard Deviation

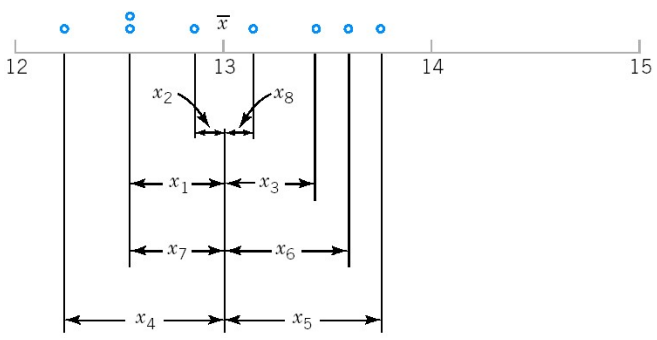
If $x_1, x_2, \ldots, x_n$ is a sample of n observations, the **sample variance** is

$$s^2 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n - 1}$$

The **sample standard deviation**, s, is the positive square root of the sample variance.

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How Does the Sample Variance Measure Variability?



How the sample variance measures variability through the deviations $x_i - \bar{x}$.

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Computation of s^2

$$s^2 = \frac{\sum_{i=1}^{n} x_i^2 - \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n}}{n-1}$$

$$V(X) = E((X-\mu)^2) = E(X^2) - E(X)^2$$

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Sample Range

If the n observations in a sample are denoted by $x_1, x_2, \ldots, x_n$, the **sample range** is

$$r = \max(x_i) - \min(x_i)$$

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Outliers

- **Outliers** are points that are much larger or smaller than the rest of the sample points.
- Outliers may be data entry errors or they may be points that really are different from the rest.
- Outliers should not be deleted without considerable thought—sometimes calculations and analyses will be done with and without outliers and then compared.

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Quartiles

Quartiles divide the data as nearly as possible into quarters.

The **first quartile** is the median of the lower half of the data.

- To find the first quartile, compute $0.25(n + 1)$. If this is an integer, then the sample value in that position is the first quartile. If not, take the average of the sample values on either side of this value.

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Quartiles

The **third quartile** is the median of the upper half of the data.

- To find the third quartile, compute $0.75(n + 1)$. If this is an integer, then the sample value in that position is the third quartile. If not, take the average of the sample values on either side of this value.

Note: The computation we used for the location of the median is equivalent to $0.5(n + 1)$. The median is the second quartile.

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Example

Suppose we have the following data:

2, 3, 5, 6, 7, 9, 9, 11, 12, 15

- What is the mean of these data?
- What is the median?
- What is the first quartile?
- What is the third quartile?

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Definition of Percentile

- The ***p*th percentile** of a sample, for a number *p* between 0 and 100, divides the sample so that as nearly as possible *p*% of the sample values are less than the *p*th percentile, and (100 − *p*%) are greater.
- The computation of the location of the *p*th percentile is analogous to what we did for the quartiles.

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To Find Percentiles

- Order the *n* sample values from smallest to largest.
- Compute the quantity $\left(\frac{p}{100}\right)(n+1)$, where *n* is the sample size.
- If this quantity is an integer, the sample value in this position is the *p*th percentile. Otherwise, average the two sample values on either side.

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Note on Percentiles

- The **first quartile** is the 25th percentile.
- The **median** is the 50th percentile.
- The **third quartile** is the 75th percentile.

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Summary Statistics for Categorical Data

The two most commonly used numerical summaries for categorical data are the **frequencies** and the **sample proportions** (sometimes called **relative frequencies**).

- The frequency for a given category is simply the number of sample items that fall into that category.
- The sample proportion is the frequency divided by the sample size.

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Summary Statistics for Categorical Data

- Example: 100 rivets are checked for their breaking strength. If 4 of the rivets fail (that is: do not hold up to the standard), find the sample proportion of rivets that fail.

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Sample Statistics and Population Parameters

- Each sample statistic in this chapter has a population parameter counterpart.
- Numerical summaries of a sample are called **statistics**.
- Numerical summaries of a population are called **parameters**.
- Statistics are often used to estimate parameters.

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Graphical Summaries

- Stem-and-leaf plot.
- Dot plot.
- Histogram.
- Boxplot.
- Scatterplot.

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Stem-and-Leaf Plot

- A simple way to summarize a data set.
- Each item in the sample is divided into two parts: a **stem**, consisting of the leftmost one or two digits, and the **leaf**, which consists of the next digit.
- It is a compact way to represent the data.
- It also gives us some indication of the shape of our data.

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Example 5

- Example: Duration of dormant periods of the geyser Old Faithful in Minutes.
- Stem-and-leaf plot:

Stem	Leaf
4	259
5	0111133556678
6	067789
7	01233455556666699
8	000012223344456668
9	013

- Let's look at the first line of the stem-and-leaf plot. This represents measurements of 42, 45, and 49 minutes.
- A good feature of these plots is that they display all the sample values. One can reconstruct the data in its entirety from a stem-and-leaf plot, although some digits may be truncated.

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Stem-and-Leaf Diagrams

Compressive Strength (in psi) of aluminum-lithium alloy specimens

105	221	183	186	121	181	180	143
97	154	153	174	120	168	167	141
245	228	174	199	181	158	176	110
163	131	154	115	160	208	158	133
207	180	190	193	194	133	156	123
134	178	76	167	184	135	229	146
218	157	101	171	165	172	158	169
199	151	142	163	145	171	148	158
160	175	149	87	160	237	150	135
196	201	200	176	150	170	118	149

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Stem-and-Leaf Diagrams

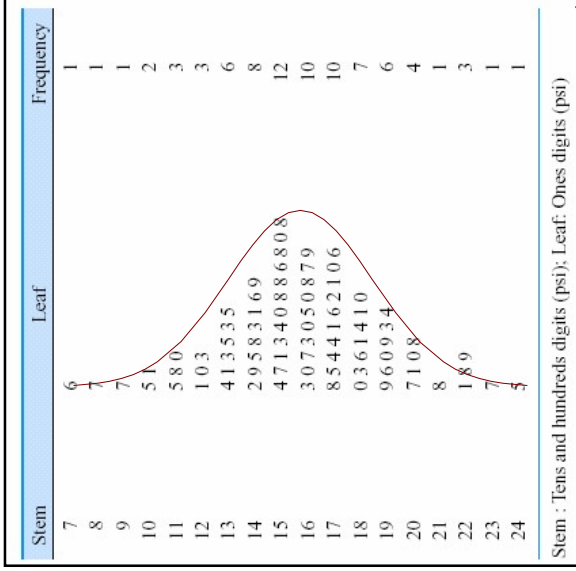
Stem	Leaf	Frequency
7	6	1
8	7	1
9	7	1
10	5 1	2
11	5 8 0	3
12	1 0 3	3
13	4 1 3 5 3 5	6
14	2 9 5 8 3 1 6 9	8
15	4 7 1 3 4 0 8 8 6 8 0 8	12
16	3 0 7 3 0 5 0 8 7 9	10
17	8 5 4 4 1 6 2 1 0 6	10
18	0 3 6 1 4 1 0	7
19	9 6 0 9 3 4	6
20	7 1 0 8	4
21	8	1
22	1 8 9	3
23	7	1
24	5	1

Stem : Tens and hundreds digits (psi); Leaf: Ones digits (psi)

Stem-and-leaf diagram for the compressive strength data

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Stem-and-Leaf Diagrams

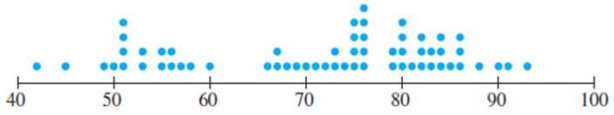


Stem-and-leaf diagram for the compressive strength data

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Dotplot

- A **dotplot** is a graph that can be used to give a rough impression of the shape of a sample, giving a good indication of where the sample values are concentrated and where the gaps are.
- It is useful when the sample size is not too large and when the sample contains some repeated values.
- It is a good method, along with the stem-and-leaf plot, to informally examine a sample. However, they are rarely used in formal presentations.
- Example (geyser data):



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Histogram

- Graphical display that gives an idea of the "shape" of the sample, indicating regions where sample points are concentrated and regions where they are sparse.
- The bars of the histogram touch each other. A space indicates that there are no observations in that interval.

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Creating a Histogram

Choose boundary points for the class intervals. Usually these intervals are the same width.

Compute the frequencies: this is the number of observations that occur in each interval.

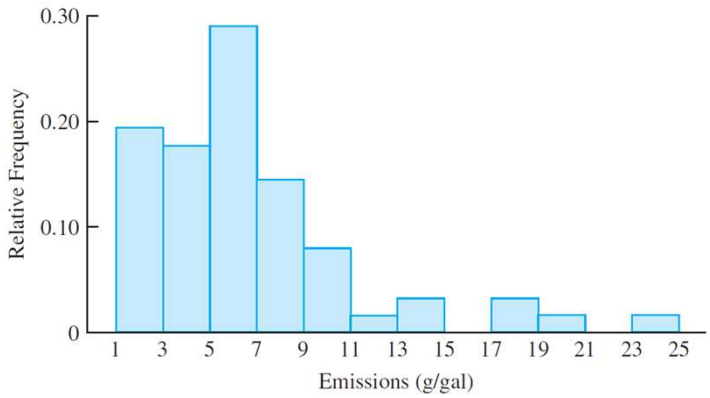
Compute the relative frequencies for each class: this is the number of observations in each interval divided by the total number of observations.

If the class intervals are the same width, then draw a rectangle for each class, whose height is equal to the frequencies or relative frequencies.

- If the class intervals are of unequal widths, the heights of the rectangles must be set equal to the densities, where **density** is the relative frequency divided by the class width.

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Example of Histogram



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Symmetry and Skewness 1

A histogram is perfectly **symmetric** if its right half is a mirror image of its left half.

- For example, the distribution of birth weights of full-term infants is roughly symmetric.

Histograms that are not symmetric are referred to as **skewed**.

A histogram with a long right-hand tail is said to be **skewed to the right**, or **positively skewed.**

- For example, the distribution of household incomes is right-skewed.

A histogram with a long left-hand tail is said to be **skewed to the left**, or **negatively skewed.**

- For example, the distribution of grades on an easy test is left-skewed.

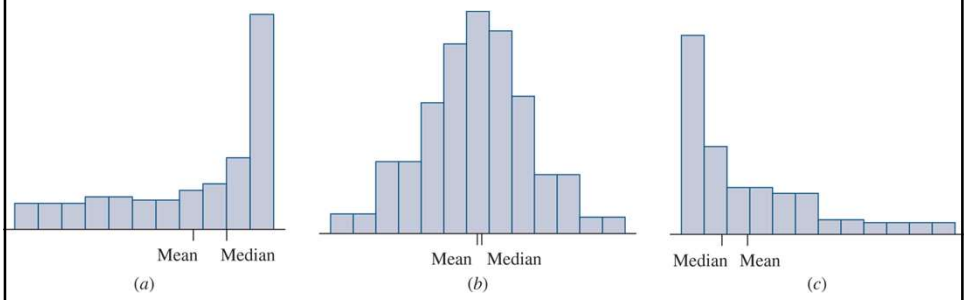
74

Symmetry and Skewness 2

- When a histogram is roughly **symmetric**, the mean and the median are approximately equal.
- When a histogram is **right-skewed**, the mean is greater than the median.
- When a histogram is **left-skewed**, the mean is less than the median.

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Skewness



Skewness<0
Left skewed
(left tail is long)

Skewness≈0

Skewness>0
Right skewed
(right tail is long)

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Unimodal and Bimodal

A histogram with only one peak is what we call **unimodal**.

If a histogram has two peaks then we say that it is **bimodal**.

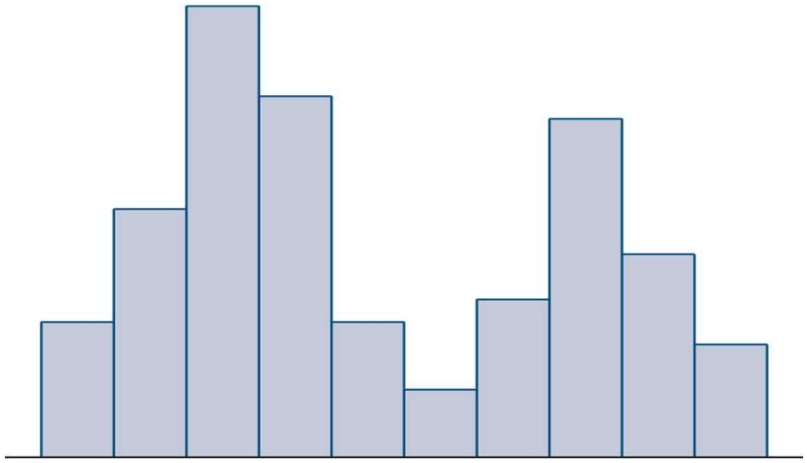
- Bimodal histograms often indicate subsamples.

If there are more than two peaks in a histogram, then it is said to be **multimodal**.

- Multimodal histograms do not often occur in practice.

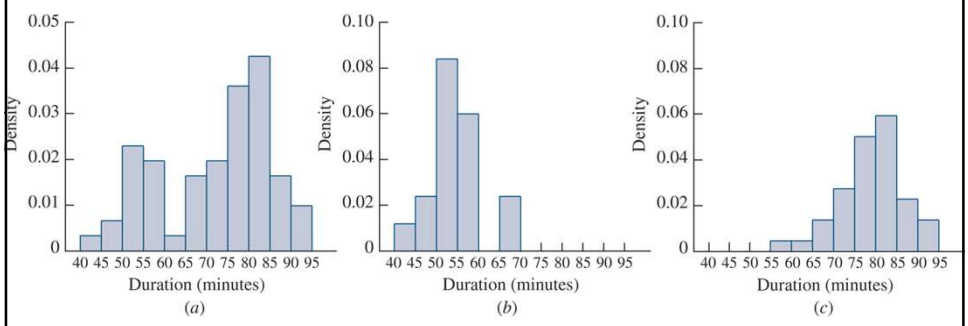
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Bimodal data



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Example



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Boxplots

- A **boxplot** is a graphic that presents the median, the first and third quartiles, and any outliers present in the sample.
- The **interquartile range (IQR)** is the difference between the third quartile and the first quartile. This is the distance needed to span the middle half of the data.

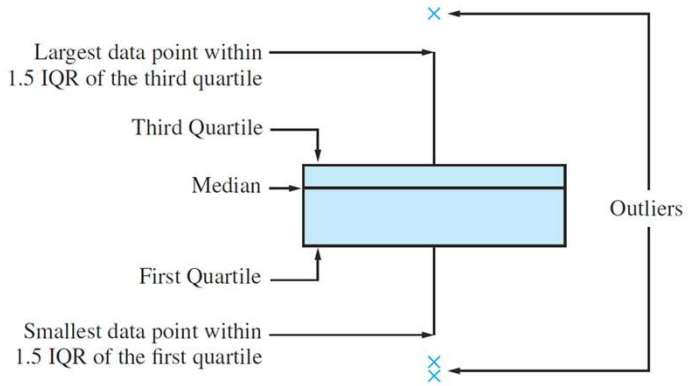
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Creating a Boxplot

- Compute the median and the first and third quartiles of the sample. Indicate these with horizontal lines. Draw vertical lines to complete the box.
- Find the largest sample value that is no more than 1.5 IQR above the third quartile, and the smallest sample value that is no more than 1.5 IQR below the first quartile. Extend vertical lines (whiskers) from the quartile lines to these points.
- Points more than 1.5 IQR above the third quartile, or more than 1.5 IQR below the first quartile, are designated as outliers. Plot each outlier individually.

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Anatomy of a Boxplot



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Class exercise

How can you explain the following statement using the notion of sampling?

"On average, your friends have more friends than you do."

What kind of assumptions (on the population and on the sample) are needed to justify this statement?

Friendship paradox - Wikipedia

The **friendship paradox** is the phenomenon first observed by the sociologist Scott L. Feld in 1991 that on average, an individual's friends have more friends than that individual.[1] It can be explained as a form of sampling bias in which people with more friends are more likely to be in one's own friend group. In other words, one is less likely to be friends with someone who has very few friends. In contradiction to this, most people believe that they have more friends than their friends have.

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